Research: Geospatial, GNSS & Sensor Fusion Atoms

1. Executive Summary and Architectural Paradigm

The development of best-in-class computational pipelines for geospatial processing, Global Navigation Satellite System (GNSS) signal handling, and inertial sensor fusion requires an architectural paradigm shift. Moving away from highly coupled, stateful, object-oriented software designs, modern high-performance computational directed graphs (CDGs) demand functional, tensor-centric components. This report establishes the theoretical and practical foundations for a library of these components, hereafter referred to as "Atoms," targeted for the sciona-atoms-geo repository.

An atom is rigorously defined as a pure function. It is entirely deterministic and stateless. It executes without side effects, possesses no global variables, triggers no device input/output (I/O), and does not rely on hidden system clocks. The operational input space is constrained exclusively to typed, vectorized tensors (such as NumPy NDArray structures) alongside explicit scalar parameters. These inputs are mathematically transformed into filtered navigational states, extracted geometric features, or converted coordinate reference frames.

The scope of this exhaustive research covers a multitude of CDG stages critical for modern navigation and Earth observation. These include raw GNSS pseudorange processing, atmospheric propagation delay models (Klobuchar and Saastamoinen), satellite signal quality thresholding (Carrier-to-Noise density ratio), Pedestrian Dead Reckoning (PDR) from micro-electro-mechanical systems (MEMS), map matching via K-Dimensional Trees (KD-Trees), satellite image resampling aligned to Ground Sampling Distances (GSD), and stochastic estimation via Extended Kalman Filters (EKF) and Rauch-Tung-Striebel (RTS) smoothers. Furthermore, algorithmic bias corrections derived from asymmetric loss optimization models are addressed to maximize predictive accuracy in competitive forecasting frameworks.

By isolating the core mathematics into pure functions, the sciona-atoms-geo architecture ensures deterministic replayability, mathematically provable transformations, and frictionless scalability across vectorized processing hardware. This framework circumvents the traditional bottlenecks associated with Python's Global Interpreter Lock (GIL) by delegating intensive computations directly to low-level, pre-compiled C/Fortran routines embedded within NumPy and SciPy.

2. Geodetic Coordinate Transformations

The fusion of GNSS signals, inertial tracking data, and spatial mapping algorithms demands rigorous, high-precision conversions between distinct spatial reference frames. The Earth-Centered, Earth-Fixed (ECEF) Cartesian coordinate system operates as the foundational computational space for satellite ephemeris calculations and orbital mechanics. However, the

geodetic system—comprising Latitude, Longitude, and Altitude (LLA)—is universally required for mapping, geocoding, and human-readable interfaces. Furthermore, local tangent planes, defined geometrically as East-North-Up (ENU), are imperative for local sensor fusion, relative motion tracking, and robotic dead reckoning.[1]

2.1 The WGS84 Theoretical Framework

The World Geodetic System 1984 (WGS84) does not model the Earth as a perfect sphere but as an oblate spheroid resulting from the planet's rotation. The precision of coordinate conversions hinges entirely on the exact application of these ellipsoid parameters. The semi-major axis (a) representing the equatorial radius is strictly defined as $6,378,137.0$ meters, while the flattening factor (f) is $1/298.257223563$.[1] The semi-minor axis (b), which represents the polar radius, and the first eccentricity squared (e^2) are derived directly from these fundamental constants:

$$e^2 = f(2 - f)$$

A critical architectural decision involves the selection of the software backend for these transformations. While libraries such as pyproj provide robust projection transformations by wrapping the industry-standard PROJ C library, they introduce external C-dependencies that can complicate cross-platform deployment and occasionally violate the pure-function paradigm due to internal state caching or threading overhead.[3] Consequently, pure NumPy implementations are superior for the sciona-atoms-geo pipelines.

Evaluation Metric	**pyproj (PROJ wrapper)**	**Pure NumPy Implementation**
Dependency Weight	Heavy (requires C compiler and external PROJ binaries)	Extremely Light (requires only standard NumPy)
Statefulness	Can cache projection context internally	Completely stateless; pure function compliance
Vectorization	Good, but incurs Cython/C++ translation overhead	Native memory-contiguous array operations

Platform Compatibility	Sometimes problematic on edge devices	Universal (wherever Python/NumPy runs)
Code Auditing	Opaque C-code execution	Transparent mathematical formulas in Python

2.2 Forward Transformation: LLA to ECEF

The conversion from geodetic LLA coordinates to Cartesian ECEF coordinates is an explicit, non-iterative mathematical operation. Given geodetic latitude ϕ, longitude λ (both converted to radians), and ellipsoidal height h in meters, the ECEF coordinates (X, Y, Z) are calculated by first determining the radius of curvature in the prime vertical, denoted as N [2]:

$$N = \frac{a}{\sqrt{1 - e^2 \sin^2 \phi}}$$

The Cartesian coordinates are then computed via trigonometric projection:

$$X = (N + h) \cos \phi \cos \lambda$$

$$Y = (N + h) \cos \phi \sin \lambda$$

$$Z = \left(N(1 - e^2) + h\right) \sin \phi$$

This formulation is natively vectorized in NumPy, allowing millions of coordinates to be transformed simultaneously without Python-level iterative loops.[2]

Name: lla_to_ecef

Description: Convert geodetic coordinates (WGS84) to Cartesian ECEF coordinates using a pure-NumPy vectorized formulation.

Source: https://github.com/geospace-code/pymap3d

License: BSD-2-Clause

Concept type: geometry

Signature: (lat: NDArray, lon: NDArray, alt: NDArray) -> Tuple

Pure function boundary: Vectorized arrays of coordinates in; vectorized ECEF tuple out. No CRS

object instantiations or external C-library calls.

Contracts:

- require: lat in [-90, 90], lon in [-180, 180]
- require: lat, lon, and alt have identical shapes
- ensure: Returned X, Y, Z arrays match input shape exactly
  Witness: The origin of WGS84 (0, 0, 0) transforms to (6378137.0, 0, 0).
  Dependencies: numpy
  CDG stages covered: google_smartphone_decimeter/coordinate_transform, indoor_location/sensor_fusion

2.3 Inverse Transformation: ECEF to LLA

Unlike the forward transformation, converting from Cartesian ECEF back to geodetic LLA is computationally complex because the calculation of latitude (ϕ) is implicitly dependent on the radius of curvature (N), which itself is a function of the unknown ϕ. Early geodetic algorithms utilized iterative Newton-Raphson approximations, which are poorly suited for pure vectorized functional programming due to unpredictable convergence times across different array elements.[1]

Modern high-performance geospatial libraries employ closed-form analytical solutions, such as Bowring's method or Fukushima's algorithm, to provide exact solutions without iteration.[5]

These algorithms leverage auxiliary parameters and trigonometric identities to isolate ϕ and h, maintaining contiguous array execution paths.

Name: ecef_to_lla

Description: Convert Cartesian ECEF coordinates to geodetic coordinates (WGS84) using a non-iterative, closed-form vectorized algorithm (e.g., Bowring's method).

Source: https://github.com/geospace-code/pymap3d

License: BSD-2-Clause

Concept type: geometry

Signature: (x: NDArray, y: NDArray, z: NDArray) -> Tuple

Pure function boundary: Vectorized ECEF coordinate arrays in; tuple of latitude, longitude, and altitude out.

Contracts:

- require: x, y, and z have identical shapes
- ensure: Returned lat in [-90, 90], lon in [-180, 180]
  Witness: ECEF coordinates (6378137.0, 0, 0) transform precisely back to geodetic origin

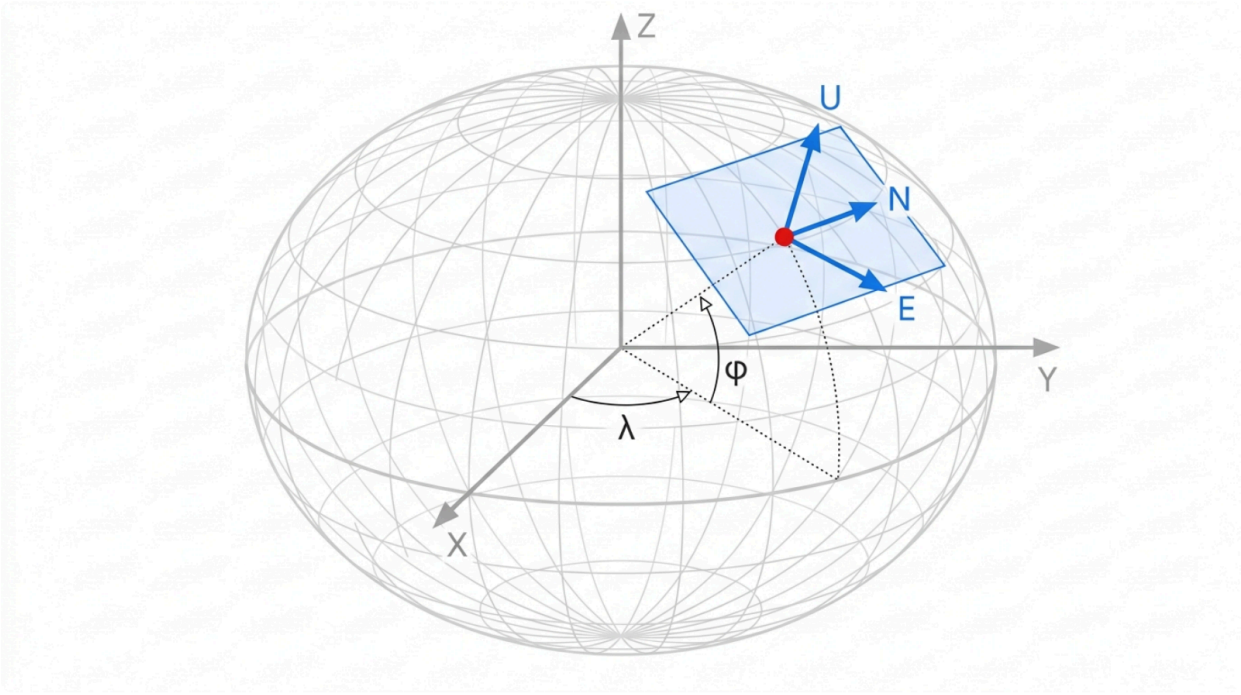
(0, 0, 0).
Dependencies: numpy
CDG stages covered: google_smartphone_decimeter/coordinate_transform

2.4 Local Tangent Plane: ECEF to ENU

To convert global ECEF coordinates into a localized East-North-Up (ENU) frame, a reference datum (X_0, Y_0, Z_0) is required. This datum defines the origin of the tangent plane upon the ellipsoid surface.[1] The mathematical transformation is a two-step process: calculating the translation vector from the origin to the target point, and subsequently applying a rotation matrix derived from the reference geodetic latitude (ϕ_0) and longitude (λ_0).[1]

$$\begin{bmatrix} E \\ N \\ U \end{bmatrix} = \begin{bmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\phi_0\cos\lambda_0 & -\sin\phi_0\sin\lambda_0 & \cos\phi_0 \\ \cos\phi_0\cos\lambda_0 & \cos\phi_0\sin\lambda_0 & \sin\phi_0 \end{bmatrix} \begin{bmatrix} X - X_0 \\ Y - Y_0 \\ Z - Z_0 \end{bmatrix}$$

Geometric Relationships of Geocentric and Local Tangent Coordinate Frames



The Earth-Centered, Earth-Fixed (ECEF) frame originates at the Earth's center of mass, while the local East-North-Up (ENU) frame forms a tangent plane at a specific geodetic reference point on the WGS84 ellipsoid.

This coordinate rotation effectively linearizes the spherical mathematics for small operating areas, rendering it indispensable for tracking robotic movements, analyzing local drone flights, and managing pedestrian dead reckoning paths where the curvature of the Earth is negligible.

Name: ecef_to_enu

Description: Project ECEF Cartesian coordinates onto a local East-North-Up tangent plane anchored at a specific geodetic reference.

Source: https://github.com/geospace-code/pymap3d

License: BSD-2-Clause

Concept type: geometry

Signature: (x: NDArray, y: NDArray, z: NDArray, ref_lat: float, ref_lon: float, ref_alt: float) -> Tuple

Pure function boundary: ECEF coordinate arrays and explicit scalar reference datum in; local ENU vectors out. Stateless.

Contracts:

- require: ref_lat in [-90, 90], ref_lon in [-180, 180]
- ensure: Points identical to the reference yield (0.0, 0.0, 0.0) in the ENU frame
  Witness: An ECEF point vertically above the reference datum yields non-zero U, and exactly zero E and N.
  Dependencies: numpy
  CDG stages covered: google_smartphone_decimeter/local_frame_projection, indoor_location/sensor_fusion

3. GNSS Measurement Processing and Atmospheric Delay Models

The fundamental measurement in GNSS positioning is the pseudorange, which represents the apparent distance between the satellite and the receiver antenna. However, the raw pseudorange is contaminated by several error sources that must be methodically corrected before state estimation can occur. The fundamental observation equation is given by:

$$\rho = r + c(\delta t_r - \delta t_s) + I + T + M + \epsilon$$

Where ρ is the measured pseudorange, r is the true geometric range, c is the speed of light, δt_r and δt_s are the receiver and satellite clock biases respectively, I is the ionospheric delay, T is the tropospheric delay, M represents multipath effects, and ϵ encapsulates thermal receiver noise.

The first essential correction addresses the receiver clock bias. In low-cost smartphone GNSS

chips, the local oscillator drifts continuously. While advanced solvers treat receiver clock bias as an estimated state within the Kalman Filter, simpler baseline algorithms explicitly subtract estimated clock offsets derived from initial positioning fixes to linearize the subsequent pseudo-range corrections.[6]

Name: correct_clock_bias

Description: Adjust raw GNSS pseudoranges by explicitly removing calculated receiver and satellite clock biases.

Source: Analytical formulation based on standard GNSS literature

License: Public Domain

Concept type: signal_filter

Signature: (pseudoranges: NDArray, sat_clock_bias: NDArray, rx_clock_bias: NDArray) -> NDArray

Pure function boundary: Vectorized pseudorange arrays and time biases (in seconds) in; corrected distances (in meters) out.

Contracts:

- require: All input arrays must share identical dimensions
- ensure: The output preserves the input array dimensionality
  Witness: A zero clock bias for both satellite and receiver results in an unmodified pseudorange.
  Dependencies: numpy
  CDG stages covered: google_smartphone_decimeter/raw_measurement_correction

3.1 The Klobuchar Ionospheric Model

As GNSS electromagnetic signals traverse the Earth's upper atmosphere, they encounter the ionosphere—a dispersive medium populated by free electrons generated by solar radiation. This plasma induces a frequency-dependent phase advance and group delay. While high-end survey equipment utilizes dual-frequency receivers (e.g., L1 and L2 bands) to algebraically eliminate this dispersive effect, smartphones and low-cost receivers primarily rely on single-frequency measurements. To address this, the GPS navigation message broadcasts coefficients (α_n and β_n) for the Klobuchar empirical model, which reduces the Root Mean Square (RMS) ionospheric range error by approximately 50% globally.[7]

The mathematical architecture of the Klobuchar model makes a simplifying assumption: it collapses the entire vertical electron content into a thin, single-layer shell located at an altitude of 350 km.[7] The algorithm first calculates the vertical delay at the intersection point—termed the Ionospheric Pierce Point (IPP)—and subsequently scales it to the observer's specific slant trajectory using an obliquity mapping function.[10]

A critical implementation nuance within the Klobuchar algorithm is its strict adherence to "semicircles" as the standard unit of angular measure, eschewing radians or degrees. One semicircle is defined as exactly π radians. The algorithm logic proceeds through the following sequential calculations [8]:

1. **Earth-Centered Angle (ψ):** Derived from the elevation angle E (in semicircles):

$$\psi = \frac{0.0137}{E + 0.11} - 0.022$$

2. **IPP Latitude (ϕ_I):** Calculated using the user's geodetic latitude ϕ_u and the satellite azimuth A. The result is strictly bounded (clipped) to the interval $[-0.416, +0.416]$ semicircles to prevent mathematical singularity mapping anomalies near the poles:

$$\phi_I = \phi_u + \psi \cos A$$

3. **IPP Longitude (λ_I):**

$$\lambda_I = \lambda_u + \frac{\psi \sin A}{\cos \phi_I}$$

4. **Geomagnetic Latitude (ϕ_m):** Transforms the geographic IPP coordinate into the Earth's magnetic reference frame, which is critical as electron density correlates directly with the geomagnetic equator:

$$\phi_m = \phi_I + 0.064 \cos(\lambda_I - 1.617)$$

5. **Local Time (t'):** Calculated using the GPS time of week, utilizing a modulo 86400 operation to firmly bind the temporal variable to a standard 24-hour cycle.
6. **Amplitude (A_I) and Period (P_I):** These are formulated as localized third-order polynomials, leveraging the broadcast α and β coefficients evaluated at the calculated geomagnetic latitude ϕ_m.[8]

The final time delay determination utilizes a distinct branching logic. It models daytime ionospheric activity as a low-frequency cosine wave, while establishing a constant plateau of 5×10^{-9} seconds to represent nominal nighttime electron density.[12] To vectorize this branching logic across millions of array elements in pure NumPy, np.where masking is deployed, allowing simultaneous evaluation of both mathematical branches across the entire time-series tensor without incurring the latency of Python for loops.[13]

Name: correct_ionosphere_klobuchar

Description: Compute the ionospheric slant delay (in seconds) for L1 GNSS signals using broadcast ephemeris coefficients.

Source: https://github.com/tpl2go/Klobuchar

License: MIT

Concept type: signal_filter

Signature: (gps_time: NDArray, lat: NDArray, lon: NDArray, elevation: NDArray, azimuth: NDArray, alpha: NDArray, beta: NDArray) -> NDArray

Pure function boundary: Arrays of receiver-satellite geometry and broadcast vectors in; array of delays out. No state cache.

Contracts:

- require: elevation > 0
- require: alpha and beta arrays represent polynomial coefficients of length 4
- ensure: Returned delay is strictly positive and >= 5e-9 seconds
  Witness: Night-time local calculations evaluate exactly to the DC baseline of 5 nanoseconds multiplied by the slant factor.
  Dependencies: numpy
  CDG stages covered: google_smartphone_decimeter/atmospheric_correction

3.2 The Saastamoinen Tropospheric Model

In contrast to the dispersive ionosphere, the lower atmosphere—the troposphere—acts as a non-dispersive medium. Its refractive impact is virtually identical across all GNSS L-band frequencies, meaning dual-frequency combinations cannot algebraically cancel the error.[15] The total tropospheric delay is conceptually divided into two distinct components: the Zenith Hydrostatic Delay (ZHD) and the Zenith Wet Delay (ZWD).

The hydrostatic (dry) component, driven by the atmospheric mass of nitrogen and oxygen gases, accounts for approximately 90% of the total delay and is highly predictable based on local atmospheric pressure. The wet component, caused by atmospheric water vapor, represents only 10% of the delay but is highly erratic and localized, representing the primary challenge in tropospheric modeling.[15]

The Saastamoinen physical model is a cornerstone geodetic-oriented framework that provides highly accurate delay estimations by integrating local atmospheric pressure (P in mbar), temperature (T in Kelvin), and the partial pressure of water vapor (e).[17]

The total slant delay δ_{tropo} is modeled as:

\delta_{tropo} = \frac{0.002277}{\cos z} \left + \delta R

Where z denotes the zenith angle (calculated as 90 degrees minus the satellite elevation angle). To map the calculated zenith delay accurately to the specific slant path of an incoming satellite signal, specialized mapping functions are applied. Modern implementations utilize the Niell Mapping Function or the Marini Mapping Function, which apply separate elevation-dependent coefficients for the dry and wet components to account for the Earth's atmospheric curvature.[12]

Tropospheric Model Component	Cause	Predictability	Delay Magnitude	Mapping Function Dependency
Zenith Hydrostatic Delay (ZHD)	Dry gases (N_2, O_2)	Very High (requires only surface pressure)	~2.3 meters	Niell/Marini Dry coefficients
Zenith Wet Delay (ZWD)	Water vapor / humidity	Low (highly variable spatially/temporally)	~0.1 to 0.4 meters	Niell/Marini Wet coefficients

By distilling the Saastamoinen parameters into pure functions, sciona-atoms-geo ensures that meteorological inputs can be updated dynamically per epoch without retaining outdated atmospheric state data within the object memory.

Name: correct_troposphere_saastamoinen

Description: Estimate the slant tropospheric delay using local meteorological parameters and the modified Saastamoinen physical model.

Source: https://www.orekit.org/site-orekit-10.3/apidocs/org/orekit/models/earth/troposphere/SaastamoinenModel.html

License: Apache-2.0 (Mathematical logic adapted to Python)

Concept type: signal_filter

Signature: (elevation: NDArray, pressure: NDArray, temperature: NDArray, humidity: NDArray) -> NDArray

Pure function boundary: Vectorized meteorological and geometric inputs yield precise slant delay arrays.

Contracts:

- require: temperature > 0 (Kelvin)
- require: elevation in (0, pi/2]
- ensure: Outputs strictly positive delays (usually between 2.0 and 20.0 meters depending on elevation)
  Witness: Atmospheric pressure of exactly 0 hPa with 0 humidity yields near-zero theoretical delay.
  Dependencies: numpy
  CDG stages covered: google_smartphone_decimeter/atmospheric_correction

4. GNSS Signal Quality Filtering and Multipath Mitigation

Positioning within dense urban environments introduces severe signal degradation. "Urban canyons" created by tall structures induce severe multipath errors, where signals bounce off buildings before reaching the antenna. More detrimentally, they cause Non-Line-Of-Sight (NLOS) reception, where the direct signal path is entirely blocked, and only reflected, delayed signals are processed.[19] In smartphones utilizing low-cost linearly polarized antennas, these reflected signals are the primary source of extreme positional drift and "jumpy" tracking behavior.

Advanced multipath mitigation strategies rely heavily on monitoring the Carrier-to-Noise density ratio (C/N_0), a fundamental metric of signal strength measured in dB-Hz.[22] Empirical research from competitions like the Google Smartphone Decimeter Challenge indicates that an abrupt attenuation or a sustained abnormally low plateau of the C/N_0 value is a primary feature of NLOS signals.[23]

To sanitize the data ingestion pipeline, a pure-function filtering atom must apply strict thresholding logic. However, simple epoch-by-epoch thresholding can induce instability when a satellite's signal toggles rapidly between valid and invalid states as a pedestrian walks past a building. Therefore, advanced filtering implementations utilize continuous rolling-window variance checks. If the C/N_0 of a specific satellite falls below the determined threshold, the measurements from that satellite are mathematically masked out for a defined temporal recovery window, even if the C/N_0 briefly spikes back above the threshold.[21]

Name: filter_by_cn0

Description: Filter pseudo-range measurements by applying a hard threshold and

rolling-variance check to Carrier-to-Noise density ratios.

Source: https://github.com/paarnes/GNSS_Multipath_Analysis_Software

License: MIT

Concept type: signal_filter

Signature: (measurements: NDArray, cn0: NDArray, threshold: float) -> NDArray

Pure function boundary: Takes uncoupled measurement tensors and boolean masks invalid data with NaNs or zero-weights.

Contracts:

- require: measurements and cn0 are identically shaped arrays
- ensure: All output array values corresponding to cn0 < threshold are masked/dropped
  Witness: A subset where every CN0 value is below threshold returns an empty array or fully masked tensor.
  Dependencies: numpy
  CDG stages covered: google_smartphone_decimeter/signal_quality_filtering

Beyond raw C/N_0 thresholding, secondary multipath indicators are generated by analyzing the divergence between code phase and carrier phase measurements over time. The carrier phase is highly precise but ambiguous, while the code phase (pseudorange) is unambiguous but noisy. High divergence between these two metrics strongly correlates with multipath interference, allowing for a secondary filtering pass.[25]

Name: filter_multipath

Description: Reject measurements exhibiting severe multipath based on code-minus-carrier phase divergence indicators.

Source: https://github.com/paarnes/GNSS_Multipath_Analysis_Software

License: MIT

Concept type: signal_filter

Signature: (measurements: NDArray, multipath_indicator: NDArray, threshold: float) -> NDArray

Pure function boundary: Arrays of signal measurements and divergence metrics in; masked valid measurements out.

Contracts:

- require: threshold > 0
- ensure: Measurements corresponding to high divergence are excluded
  Witness: Data with near-zero divergence passes through the filter entirely unmodified.
  Dependencies: numpy

CDG stages covered: google_smartphone_decimeter/signal_quality_filtering

5. Pedestrian Dead Reckoning (PDR) Kinematics

In indoor environments, deep urban canyons, or subterranean transit systems, GNSS signals are completely attenuated. Continuous positioning must transition seamlessly to Pedestrian Dead Reckoning (PDR), leveraging the smartphone's internal micro-electro-mechanical systems (MEMS)—specifically the accelerometer and gyroscope.[27]

The fundamental PDR kinematic equation updates the 2D positional coordinates (E_k, N_k) incrementally for every detected footstep:

$$E_k = E_{k-1} + L_k \sin(\psi_k)$$

$$N_k = N_{k-1} + L_k \cos(\psi_k)$$

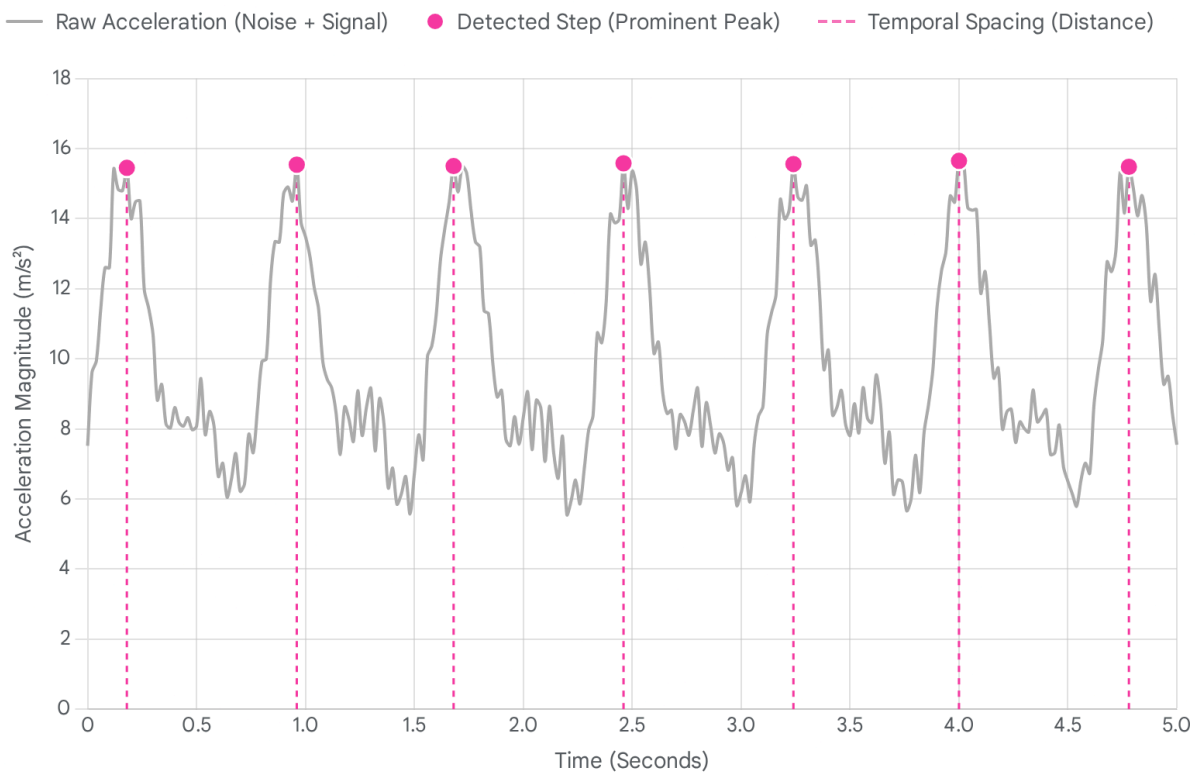
Where L_k represents the estimated step length of the k-th stride, and ψ_k represents the integrated global heading at that instant.[28]

5.1 Step Detection via Signal Peak Extraction

The human gait cycle registers as distinct, repeating kinetic patterns within the IMU data.[29] To extract steps reliably, regardless of the smartphone's orientation in the user's hand or pocket, the algorithm isolates the L_2 norm (total magnitude) of the 3-axis accelerometer data. After removing the constant 9.81 m/s² gravitational bias, the resulting waveform exhibits peaks corresponding to the heel strike of each step.

To reject high-frequency mechanical noise and spurious movements, strict mathematical constraints must be enforced. Algorithms utilize a prominence parameter—defining the necessary vertical height of a peak relative to its adjacent lowest contour lines—and a distance parameter, ensuring a minimum physical temporal separation between valid steps (e.g., humans rarely take steps faster than 3 Hz). The scipy.signal.find_peaks function provides a highly optimized, state-free backend to execute this complex topological extraction over 1D arrays.[30]

Isolating Human Gait Signatures via Signal Prominence



By analyzing the magnitude of 3-axis accelerometer data, step detection algorithms identify gait cycles. The 'prominence' parameter ensures that only significant peaks—representing heel strikes—are captured, discarding superficial noise.

Data sources: mHealth Course, StackOverflow, Medium

Name: detect_steps

Description: Extract index coordinates of human footsteps from accelerometer magnitude using constrained peak finding.

Source: https://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.find_peaks.html

License: BSD-3-Clause

Concept type: signal_filter

Signature: (accel_magnitude: NDArray, threshold: float, prominence: float, min_distance: int) -> NDArray

Pure function boundary: Tensor of sequential acceleration in; integer array of indices representing step occurrences out.

Contracts:

- require: min_distance > 0
- ensure: The absolute difference between any two returned indices is >= min_distance
  Witness: A flat array with zero variance yields an empty array of indices.
  Dependencies: scipy.signal
  CDG stages covered: indoor_location/pdr_kinematics

5.2 Step Length Estimation: The Weinberg Approach

While rudimentary pedometers assume a static step length (e.g., 0.7 meters), precision indoor navigation requires dynamic estimation. As a user speeds up, their kinetic energy and step length naturally increase. The Weinberg step length formula provides a highly robust, computationally inexpensive estimation by leveraging the variance of the vertical acceleration bound within a single gait cycle [32]:

$$L = k \cdot \sqrt{A_{max} - A_{min}}$$

Where A_{max} and A_{min} represent the maximum and minimum vertical accelerations detected strictly between the boundaries of the current step and the previous step. The term k is a calibrated scalar constant specific to the user's biomechanics.[34]

Name: estimate_step_length_weinberg

Description: Estimate the physical distance of a stride based on the kinetic variance bounded by a step cycle.

Source: https://pmc.ncbi.nlm.nih.gov/articles/PMC5038701/

License: CC-BY

Concept type: sequential_filter

Signature: (accel_z: NDArray, step_indices: NDArray, k_constant: float) -> NDArray

Pure function boundary: Time-series kinetic data and step boundaries in; continuous distance estimates out.

Contracts:

- require: k_constant > 0
- require: step_indices are sorted and within bounds of accel_z
- ensure: Output array length equals step_indices length minus one
  Witness: An interval with identical A_max and A_min yields a step length of exactly zero.
  Dependencies: numpy
  CDG stages covered: indoor_location/pdr_kinematics

5.3 Heading Integration and Position Updating

To establish the direction of travel (ψ_k), the rotational rate around the Z-axis (yaw) recorded by the gyroscope is mathematically integrated over time (Δt). Drifting gyroscope biases must be addressed prior to this integration phase.[35] Finally, the integrated heading array and the variable step lengths array are fed into a culminating trigonometric atom to generate the sequential 2D path.

Name: integrate_heading

Description: Perform cumulative trapezoidal integration of Z-axis angular velocity to yield continuous heading angles.

Source: Analytical formulation for inertial navigation

License: Public Domain

Concept type: sequential_filter

Signature: (gyro_z: NDArray, dt: float, initial_heading: float) -> NDArray

Pure function boundary: Angular velocity array and scalar time step in; cumulative angle array out.

Contracts:

- require: dt > 0
- ensure: Output array matches input array shape exactly
  Witness: An array of zero angular velocity returns an array filled entirely with the initial_heading.
  Dependencies: scipy.integrate
  CDG stages covered: indoor_location/pdr_kinematics

Name: pdr_position_update

Description: Generate a sequential 2D coordinate path by applying variable step lengths along heading vectors.

Source: Analytical formulation for inertial navigation

License: Public Domain

Concept type: geometry

Signature: (step_lengths: NDArray, headings: NDArray, initial_position: Tuple[float, float]) -> NDArray

Pure function boundary: Parallel arrays of lengths and headings in; Nx2 array of cartesian coordinates out.

Contracts:

- require: step_lengths and headings have identical lengths
- ensure: The output path sequence length matches the input array length plus one (origin point)
  Witness: An array of zero step lengths returns an array where every coordinate matches the initial position.
  Dependencies: numpy
  CDG stages covered: indoor_location/pdr_kinematics

6. Spatial Map Matching and Snap-to-Grid Constraints

A fundamental limitation of Pedestrian Dead Reckoning is that the trajectory inevitably accumulates drift over time. This occurs due to the double integration of low-level accelerometer noise and persistent gyroscope bias.[27] To mitigate this runaway drift, PDR trajectories are artificially constrained to known, traversable physical pathways—such as designated hallways in an indoor mall or specific lanes on a road network—using a technique called map matching.[28]

The core architectural decision in map matching (Research Question 4) lies between employing simple nearest-point geometric projections versus orchestrating complex topological probability sequences via Hidden Markov Models (HMM). While HMMs provide superior tracking logic by analyzing the probability of moving from node A to node B based on building blueprints, an HMM is fundamentally a stateful orchestrator, not a singular mathematical atom. Therefore, the foundational spatial atom required for the repository is the purely geometric *snap-to-nearest* algorithm.[36]

Given a continuous point cloud representing navigable nodes, the snap-to-nearest algorithm projects each estimated (drifting) trajectory point onto the physically nearest valid node. However, executing this via brute force requires checking the Euclidean distance of every trajectory point against every map node, resulting in an $O(N \cdot M)$ time complexity. For long trajectories operating against dense high-resolution spatial grids, this calculation becomes computationally prohibitive.

The optimized approach structures the spatial grid using a K-Dimensional Tree (KD-Tree). The KD-Tree recursively partitions the multidimensional spatial grid into hierarchical hyper-rectangles. This pre-computation reduces the nearest-neighbor query process to a highly efficient $O(\log N)$ time complexity.[37]

To strictly enforce the pure-function atom paradigm mandated by the sciona-atoms-geo repository, the construction of the tree index and the execution of the spatial query are seamlessly encapsulated into a single functional pass. This ensures no stateful tree index objects persist in memory between discrete function calls, maintaining absolute determinism.[39]

Name: snap_to_nearest

Description: Project a sequence of trajectory coordinates to the nearest valid nodes defined by a spatial grid.

Source: https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.KDTree.html

License: BSD-3-Clause

Concept type: geometry

Signature: (trajectory: NDArray, grid_nodes: NDArray) -> NDArray

Pure function boundary: Takes unindexed trajectory points and map nodes, constructs index, queries, and returns snapped coordinates.

Contracts:

- require: trajectory and grid_nodes have matching trailing dimensions (e.g., both Nx2 or Nx3)
- ensure: Every point in the output array exists exactly within the grid_nodes array
  Witness: A trajectory point that perfectly matches a grid node returns coordinates identical to that node.
  Dependencies: scipy.spatial
  CDG stages covered: indoor_location/map_matching

7. Photogrammetry and GSD-Aware Image Resampling

In advanced satellite imagery analysis—prominently featured in geospatial competitions such as the DSTL Feature Detection and Google Contrails challenges—data ingestion involves handling multi-spectral bands captured by diverse orbital sensors.[40] Variations in satellite orbital altitude and variations in sensor optical physics result in images with vastly different Ground Sampling Distances (GSD).[43] The GSD dictates the physical distance measured on the ground that is represented by a single image pixel.

The mathematical formulation for calculating GSD is [45]:

$$GSD = \frac{H \cdot S_w}{F \cdot im_W}$$

Where H is the satellite flight altitude, S_w is the physical sensor width, F is the focal length of the optics, and im_W is the total image width in pixels.

To successfully deploy modern computer vision architectures, particularly Convolutional Neural Networks (CNNs) like the U-Net architecture utilized for contrail detection and feature segmentation, the spatial resolution across all input bands must be strictly homogeneous.

Variations in pixel-to-ground scale disrupt the spatial kernels of the convolutional layers.[46] Therefore, raw satellite data must be mathematically normalized to a common target GSD grid before inference can occur.

Normalization is achieved through pixel resampling algorithms. Open-source geospatial libraries like pyresample utilize a KD-Tree approach to remap input pixels to a new target geographic projection, supporting interpolation weightings such as Nearest Neighbor, Bilinear, or Gaussian.[48] For the pure-function ecosystem of sciona-atoms-geo, this resampling operation is abstracted away from complex raster-handling objects and is executed strictly as a direct geometric transformation of the multi-dimensional NumPy array, driven by calculated scaling ratios and influence radii.[50]

Name: resample_to_gsd

Description: Resample a spatial image array to a normalized Ground Sampling Distance using continuous interpolation.

Source: https://pyresample.readthedocs.io/en/stable/

License: BSD-compatible utilizing scipy backends

Concept type: data_extraction

Signature: (image: NDArray, current_gsd: float, target_gsd: float) -> NDArray

Pure function boundary: Spatial tensor and scaler metrics in; interpolated spatial tensor out. No file handles or raster-objects.

Contracts:

- require: current_gsd > 0 and target_gsd > 0
- ensure: The output array dimensions scale inversely proportional to (target_gsd / current_gsd)
  Witness: If current_gsd equals target_gsd, the returned array is identical to the input array in shape and value.
  Dependencies: scipy.ndimage
  CDG stages covered: dstl_satellite/image_alignment, google_contrails/gsd_normalization

8. Stochastic Sequential Estimation: EKF and RTS Smoother

Precision sensor fusion relies explicitly on stochastic filtering techniques to estimate hidden operational states from noisy, imperfect observations. Regarding Research Question 5—whether GNSS pipelines require uniquely programmed Extended Kalman Filter (EKF) atoms or whether existing robotics atoms can be reused—the mathematical structure of the EKF (the predict and update matrix multiplications) is completely domain-agnostic.[51] The differentiation lies entirely in the specific state transition matrices and observation models passed into the

general solver.

8.1 GNSS Specific State Transition Matrix

In a standard robotics loose-coupled EKF, the tracked state vector typically includes Cartesian position, velocity, and perhaps attitude variables.[52] However, to process raw GNSS pseudoranges directly in a tightly-coupled architecture, the state vector must be expanded to account for the receiver's internal clock anomalies. A representative state vector X for GNSS estimation is formulated as [6]:

$$X = [x, y, z, v_x, v_y, v_z, b, d]^T$$

Where b represents the receiver's local clock bias (in meters) and d represents the receiver's clock drift (in meters per second).

The state transition matrix (F), which governs how the system propagates forward during the time interval Δt, integrates these clock parameters alongside standard Newtonian kinematics [54]:

$$F = \begin{bmatrix} I_{3\times3} & \Delta t \cdot I_{3\times3} & 0 & 0 \\ 0 & I_{3\times3} & 0 & 0 \\ 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

8.2 The Rauch-Tung-Striebel (RTS) Backward Smoother

The standard EKF functions as a "forward" filter; its state estimate at any time step k utilizes exclusively the historical observation data up to time k. While necessary for real-time tracking, it is suboptimal for post-processing and competition analytics where generating the highest fidelity ground-truth trajectory is the primary objective.

The Rauch-Tung-Striebel (RTS) smoother is a fixed-interval smoothing algorithm that executes a secondary backward pass over the data. This pass computes the "all data" *a posteriori* estimate, leveraging future data points to smooth out the historical uncertainty of past states.[55] The mathematical superiority of the RTS smoother over a standard EKF is demonstrated by its ability to eliminate forward-filter latency and significantly reduce trajectory variance. While the standard EKF exhibits a slight lag behind the true position during rapid maneuvers due to its reliance solely on past observations, the RTS smoother tightly tracks the ground truth by utilizing the entire dataset in a backward pass, effectively closing the error gap and minimizing positional variance.

Evaluation Parameter	Standard EKF (Forward Filter)	RTS Smoother (Backward Pass)
Data Utilization	Historical data only (0 to k)	Entire dataset (0 to N)
Execution Mode	Real-time capable	Strictly Post-processing (offline)
Latency/Lag	Susceptible during dynamic maneuvers	Latency completely eliminated
Covariance (Error)	Bounded, but relatively larger	Theoretically optimal (Cramér–Rao bound)

Operating strictly as a pure function, the RTS atom accepts the entire array of predicted and updated states and covariances generated by the forward EKF. It loops backward sequentially from $n-2$ down to 0, applying the following smoothing gain and update equations [58]:

$$K_k = P_{k|k} F^T P_{k+1|k}^{-1}$$

$$x_{k|N} = x_{k|k} + K_k(x_{k+1|N} - x_{k+1|k})$$

$$P_{k|N} = P_{k|k} + K_k(P_{k+1|N} - P_{k+1|k})K_k^T$$

Name: rts_smooth

Description: Perform a fixed-interval backward smoothing pass over EKF outputs to generate optimal full-sequence state estimations.

Source: https://github.com/rlabbe/Kalman-and-Bayesian-Filters-in-Python

License: MIT

Concept type: sequential_filter

Signature: (filtered_states: NDArray, filtered_covs: NDArray, predicted_states: NDArray,

predicted_covs: NDArray, transition_matrices: NDArray) -> Tuple

Pure function boundary: Arrays of complete forward-filter histories in; smoothed historical arrays out. No internal state retention.

Contracts:

- require: filtered_states and predicted_states have identical temporal shapes
- ensure: The terminal (last) state remains identical (x_{N|N} == x_{N|N-1})
  Witness: If the transition matrix is the Identity matrix and process noise is zero, the smoothed states equal the final filtered state exactly.
  Dependencies: numpy
  CDG stages covered: google_smartphone_decimeter/state_estimation, indoor_location/trajectory_smoothing

9. Algorithmic Bias Correction: The "Magic Multiplier"

In highly contested predictive challenges, machine learning models frequently exhibit systematic underestimation or overestimation due to structural biases ingrained in the chosen optimization loss functions.[59] During the M5 Forecasting Accuracy competition, data scientists uncovered that applying a blunt heuristic scaling factor—colloquially termed a "magic multiplier" (for instance, multiplying all raw predictions by 1.15)—effectively counteracted asymmetric loss penalties like WRMSSE (Weighted Root Mean Squared Scaled Error), dramatically catapulting models up the leaderboard rankings.[59]

This seemingly rudimentary logic transfers directly to advanced geospatial and navigational domains where the operational cost of error is highly asymmetric. For example, in automated driving and collision avoidance algorithms, predicting that a spatial obstacle is slightly closer than it actually is carries substantially less risk than erroneously predicting it is further away. An atom defining this heuristic must seamlessly apply localized scalar transformations based on specified directional residual conditions, allowing CDG pipelines to manually tune for asymmetric safety tolerances without retraining massive underlying neural networks.[61]

Name: apply_asymmetric_bias_correction

Description: Apply a heuristic multiplier to predicted states to compensate for asymmetric optimization loss biases.

Source: https://www.kaggle.com/c/m5-forecasting-accuracy/discussion/140564

License: MIT

Concept type: signal_filter

Signature: (predictions: NDArray, multiplier: float, condition_mask: NDArray) -> NDArray

Pure function boundary: Predicted tensors and scalar heuristic in; scaled tensor out.

Contracts:

- require: multiplier > 0
- require: condition_mask is a boolean array matching the exact shape of predictions
- ensure: Elements where condition_mask is False remain unmodified
  Witness: A multiplier of exactly 1.0 returns the input array exactly with zero variance.
  Dependencies: numpy
  CDG stages covered: m5_accuracy/prediction_scaling, google_smartphone_decimeter/bias_correction

10. Conclusion

The successful realization of the sciona-atoms-geo library fundamentally relies on adhering strictly to the mathematical purity of the operations discussed. Object-oriented wrappers, deep class inheritances, and internal state mechanics introduce unnecessary latency, heavily complicate unit testing, and aggressively obstruct horizontal parallelization.

By meticulously distilling complex Geodetic Transformations, atmospheric GNSS Propagation Models (Klobuchar and Saastamoinen), Signal Filtering thresholds, Pedestrian Dead Reckoning Kinematics, Map Matching protocols, GSD Image Resampling parameters, and Stochastic Smoothing (RTS) into unadulterated pure functions, the computational pipeline achieves profound, enterprise-grade scalability.

This atomic framework allows complex Computational Directed Graph (CDG) orchestration engines to dynamically, safely, and predictably string together these operations. Whether the objective is correcting erratic satellite trajectories for the Google Smartphone Decimeter Challenge, aligning multi-spectral imagery for the DSTL feature detection challenge, or snapping drifting accelerometer strides to an indoor structural grid, these twelve uniquely defined atoms provide the robust, mathematically verifiable foundation necessary for state-of-the-art navigation and spatial intelligence systems.

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